

2025
(Session : 2024-26)

Time : 3 hours

Full Marks : 70

Pass Marks : 32

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer any **five** questions in which*

Q. No. 1 is compulsory.

1. Answer any **four** questions of the following :

$$3\frac{1}{2} \times 4 = 14$$

- (a) What are the significance of quantum numbers n , l , and m associated with the H-atom ?
- (b) State Lande's Interval Rule.
- (c) Find the spectroscopic term of $p^1 d^1$.

- (d) What are the important features of vibrational spectra ?
- (e) Why are semiconductor lasers also called Diode lasers ?
- (f) What do you mean by Second Harmonic Generation ?
2. Derive an expression for the spin-orbit interaction energy for a single electron system. 14
3. Define Larmor precession. Derive the Larmor precession frequency for an electron. $4+10 = 14$
4. Explain Normal and Anomalous Zeeman effects in detail. 14
5. Explain the rotational vibrational spectra of a diatomic molecule. 14
6. Discuss the Laser-rate equations for a three-level laser system. 14
7. Describe the experimental setup of an NMR spectrometer and explain the function of each component. 14

8. Explain the construction and working of Ruby Laser. 14

9. Write short notes on any **two** of the following :

7×2 = 14

- (a) S. G. Experiment
- (b) LS and J-J couplings
- (c) X-ray emission and absorption
- (d) Symmetric and Antisymmetric wave functions

